Mid

Lecture 1

1. Define Operating System with example.

Ans: An operating system (OS) is a software that manages a computer's hardware and software resources, acting as an intermediary between the user and the computer's hardware. Example- Microsoft Windows, Apple's macOS, and Linux.

2. Explain Computer System Structure.

Ans: Computer system can be divided into four components:

- **Hardware –** provides basic computing resources- CPU, memory, I/O devices
- **Operating system -** Controls and coordinates use of hardware among various applications and users.
- **Application programs –** Word processors, compilers, web browsers, database systems, video games
- **Users-** People, machines, other computers.

3. Four Component of a Computer System.

Ans: page 4.

4. i. Define Bootstrap program.

Ans: The initial program that runs when a computer is powered up or rebooted.

→It is stored in the ROM
→It must locate and load into memory the OS Kernel.

ii. Define Interrupt.
Ans: Interrupt is a signal that tells the CPU something important has happened and it needs to stop what it's doing and pay attention.

iii. Define System Call/Monitor Call.
Ans: Software may trigger an interrupt by executing a special operation called System Call.

5. Operating System goals/Objectives.
Ans: • Make the computer system convenient to user

- Use the computer hardware in an efficient manner

6. Function/Characteristics of Operating System.
Ans:
- **Process Management**: Create, schedule (start/paus/terminate), run, sync, communicate, and end processes.
- **Memory Management**: Track, allocate, free, and protect memory.
- **I/O Device Management**: Control devices, manage data flow, handle errors, and allocate devices.
- **File Management**: Create, organize, access, protect, and back up files.
- **Virtual Memory**: Extend RAM to run large programs using paging and swapping.
- **Security**: Protect data and resources with authentication, authorization, and encryption.

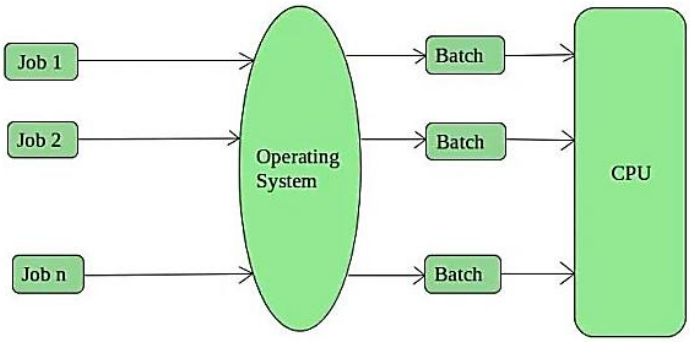
Lecture 2

7. Types of Operating System and explain any three types of OP.

Ans:
- Batch Operating System
- Multi-Programming System
- Multi-Processing System
- Multi-Tasking Operating System
- Time-Sharing Operating System
- Distributed Operating System
- Network Operating System
- Real-Time Operating System

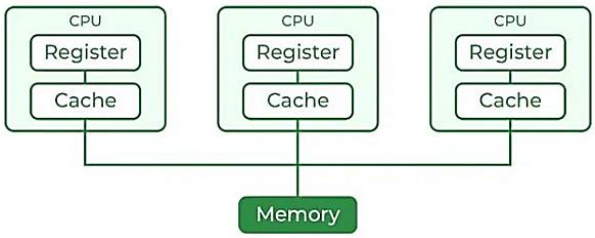
i. Batch Operating System:

Here the operator takes similar jobs having the same requirements and groups them into batches. Batch Operating System is designed to manage and execute a large number of jobs efficiently by processing them in groups.



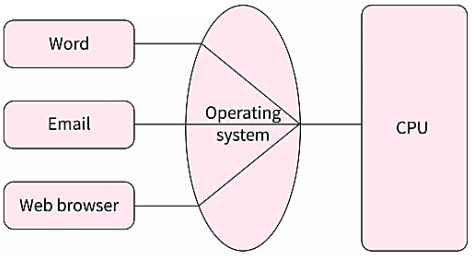
ii. Multi-Processing Operating System:

Multi-Processing Operating System is a type of Operating System in which more than one CPU is used for the execution of resources. It betters the throughput of the System.



iii. Multi-Tasking / Time Sharing Operating System:

Multitasking Operating System is simply a multiprogramming Operating System with having facility of a Round-Robin Scheduling Algorithm. It can run multiple programs simultaneously.



8. Define Uniprocessor.

Ans: **Uniprocessor** is a computer system type that has only one central processing unit which means that only one instruction will be executed at a time.

Lecture 3

9. Define Process Concept.

Ans: i. Process is a program in execution; an active program can be called a process.

ii. Textbook uses the terms ***job*** and ***process*** almost interchangeably

iii. Execution of program started via GUI mouse clicks, command line entry of its name, etc.

10. Program vs Process.

Program	Process
1. It is a passive entity.	1. It is an active entity.
2. It requires memory space to store instructions.	2. It requires resources such as CPU, memory address, I/O during its working.
3. It doesn't have a control block.	3. It has its own control block
4. It resides in the secondary memory of the system.	4. Process is being loaded into the main memory.

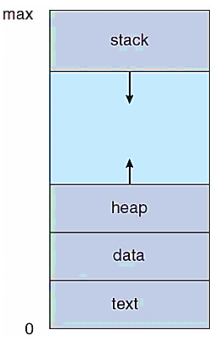
11. Explain Process in Memory.

Ans: **i. Stack:** Containing temporary Data function Parameters, return address, local variable.

ii. Heap: Containing Memory dynamically allocated during run time.

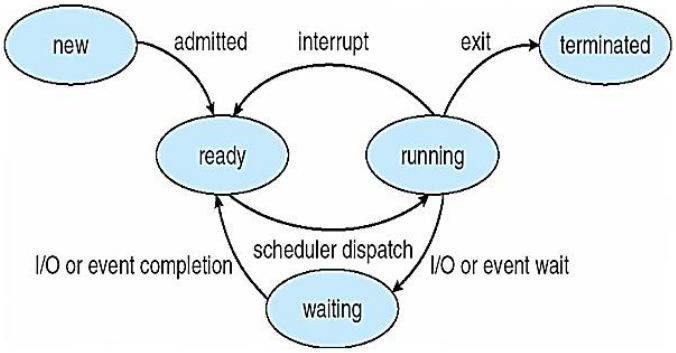
iii. Data section: Containing global and static variables.

iv. Text Section: The program code also called text section.



Process in Memory

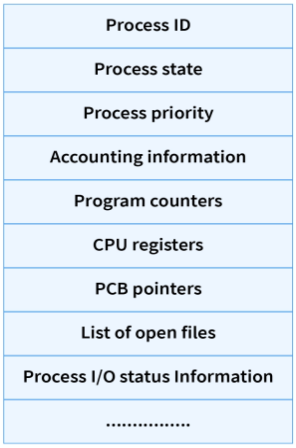
12. Process State With diagram.



Ans: As a process executes, it changes state

– **new**: The process is being created
– **running**: Instructions are being executed
– **waiting**: The process is waiting for some event to occur
– **ready**: The process is waiting to be assigned to a processor
– **terminated**: The process has finished execution

13. Explain Process Control Block (PCB).

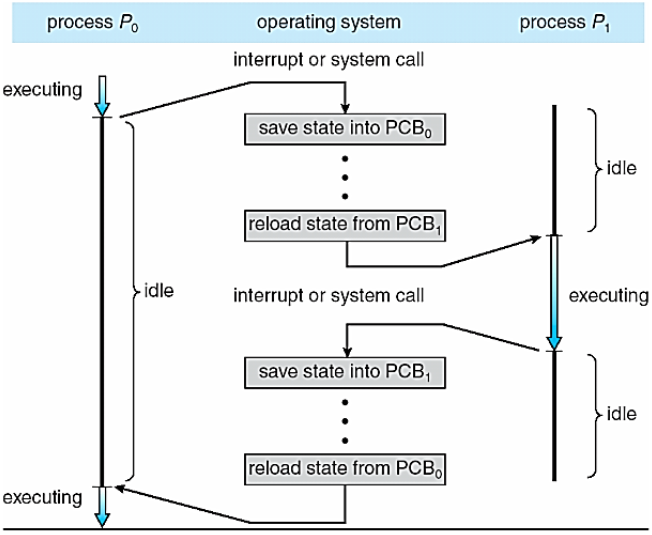


Ans:

- **Process ID:** When a new process is created by the user, the operating system assigns a **unique ID.**
- **Process State:** New, running, ready, waiting etc.
- **Program Counter:** Location of instruction to next execute.
- **Process Priority:** Process priority is a numeric value that represents the priority of each process.
- **CPU register:** Contents of all process.
- **PCB Pointer:** This maintains an easy control flow between parent processes and child processes.

14. Define Context Switching and explain with diagram.

Ans: Context switching in an operating system (OS) refers to the process where the CPU switches from executing one process to another.



The steps involved in context switching are as follows -

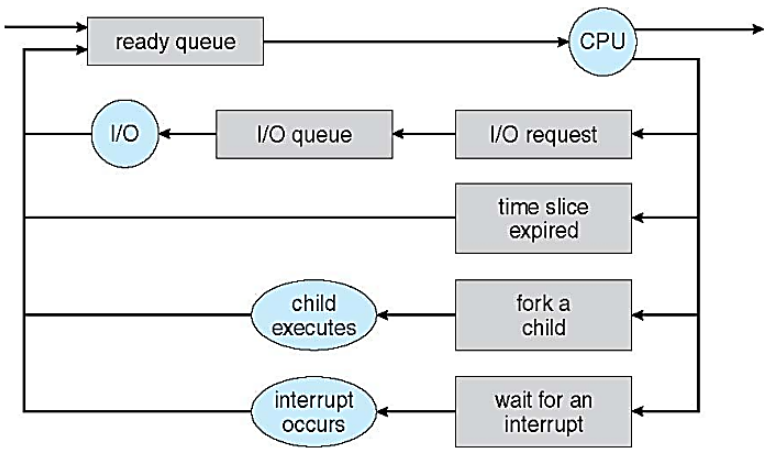
• Save the context of the process that is currently running on the CPU.
• Select a new process for execution.
• Update the memory management data structures as required.
• Update the process control block of the selected process.

15. Maintains Scheduling queues of Process and Representation of Process Scheduling.

Ans: – **Job queue:** Set of all processes in the system.
– **Ready queue:** Set of all processes residing in main memory, ready and waiting to execute.
– **Device queues:** Set of processes waiting for I/O device processes migrate among the various queues

Representation of Process Scheduling:

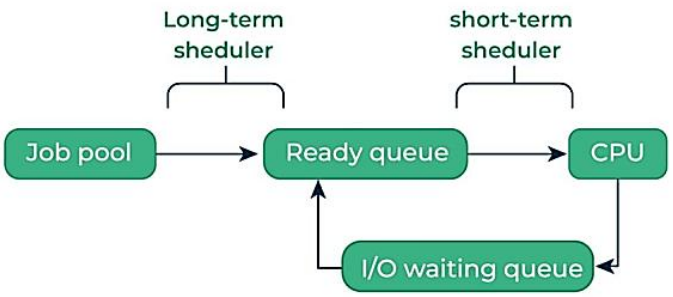
Queueing diagram represents queues, resources, flows



Lecture 4

16. Explain Long term scheduler / Job Scheduler and Sort term Scheduler/ CPU Scheduler with diagram.

Ans: i. Long-term scheduler (or **job scheduler**) –

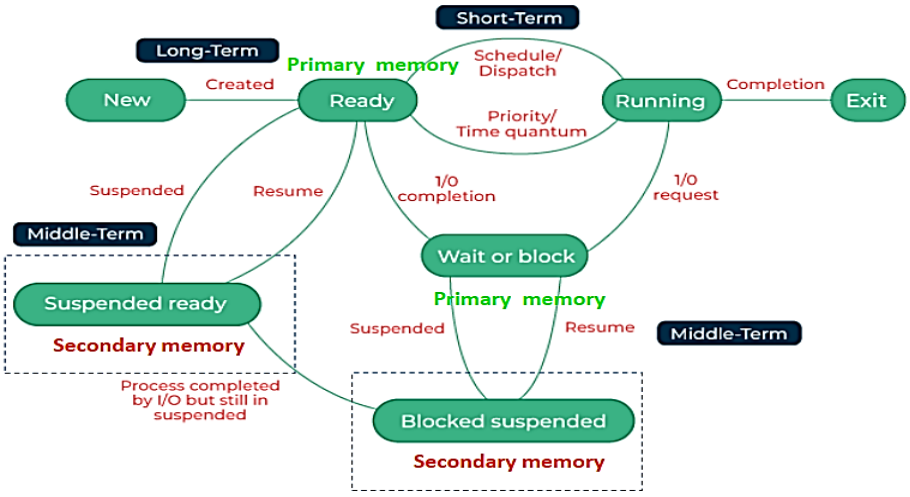


- It mainly moves processes from Job Queue to Ready Queue.
- It controls the Degree of Multi-programming
- It is important that the long-term scheduler make a careful selection of both I/O and CPU-bound processes.

ii. **Short-term scheduler (or CPU scheduler)**

- STS (Short Term Scheduler) must select a new process for the CPU frequently to avoid starvation.
- It picks a process from ready queue.
- It mainly calls dispatcher.
- Its main objective is to make the best use of CPU.

17. Full diagram of Process Schedulers.



18. Differentiate Preemptive and Non-Preemptive.

Preemptive	Non-Preemptive
In these resources (CPU Cycle) are allocated to a process for a limited time	Once resources (CPU Cycle) are allocated to a process
Process can be interrupted in between.	Process can not be interrupted until it terminates itself
flexible	Rigid
Preemptive scheduling waiting time is less	Non-preemptive scheduling waiting time is high
Cost associated	No cost associated
Preemptive scheduling response time is less	Non-preemptive scheduling response time is high

19. Different time Concepts in CPU Scheduling.

i. Arrival Time: In CPU Scheduling, the arrival time refers to the moment in time when a process enters the ready queue and is awaiting execution by the CPU.

ii. Burst Time: Burst time, also referred to as "execution time". It is the amount of CPU time the process requires to complete its execution.

iii. Completion Time: Completion time is when a process finishes execution and is no longer being processed by the CPU. It is the summation of the arrival, waiting, and burst times.

iv. Turnaround Time: Turnaround Time = Completion Time – Arrival Time

v. Waiting Time: Waiting Time = Turnaround Time – Burst Time

vi. Response Time: Response Time = Time it Started Executing – Arrival Time

Lecture 5

20. What is First Come First Serve (FCFS) Scheduling?

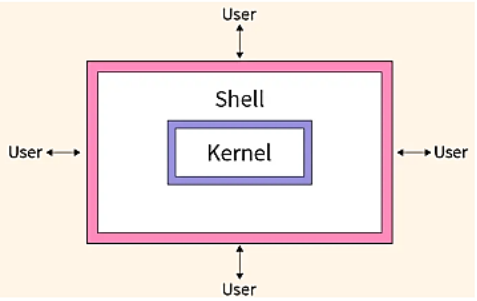
Ans: The first come first serve scheduling algorithm states that the process that requests the CPU first is allocated the CPU first. It is implemented by using the FIFO queue. FCFS is a non-preemptive scheduling algorithm.

21. Example Math.

Ans: Page 3 to 4

Lab Lecture 1

22.Define Kernel and Shell with diagram.



Ans: Kernel: The **kernel** is the heart of the operating system, responsible for direct communication with the hardware. It manages critical functions such as controlling the CPU, handling memory, and ensuring that multiple processes.

Shell: A shell is a program that provides an interface between a user and an operating system (OS) kernel.

23. Define Command line interpreter.

Ans: Shell is a UNIX term for the interactive user interface with an operating system. In some systems, the shell is called a command line interpreter.

24. Define Shell Script.

Ans: A shell script is a program composed of a series of operating system commands that are executed by the command-line interpreter (CLI) or the shell in sequence.

25. What is Base Script?

Ans: Bash is a Unix command line interface responsible for interacting with a computer's operating system. **a bash script is a useful way to group commands to create a program**.